Fiscal Year:	FY 2004	**Task Last Updated:**	FY 02/02/2005
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Project Title:	Mechanisms by which Selenomethionine Protects Against Space Radiation Biological Effects		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
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Task Description:	This research program is focused on the mechanism(s) by which L-selenomethionine reduces the levels of biological effects associated with the exposure of cells to high-energy, highly charged particles known as HZE particles from the new NASA BAF facility at the Brookhaven National Laboratory. It is known that L-selenomethionine can reduce the levels of oxidative stress produced by cellular irradiation with x-rays, protons and HZE particles. It is expected that L-selenomethionine will serve as an effective countermeasure for other types of space radiation induced biological effects, such as the induction of cancer. The studies being performed as part of this grant are exploring the mechanisms by which L-selenomethionine can serve as a countermeasure for radiation effects, including the mechanisms by which it may regulate gene expression related to radiation induced carcinogenesis. We hypothesize that L-selenomethionine supplementation triggers expression and accumulation of ATR and CHK2 kinases, that in turn regulate early signaling elements in checkpoint pathways, namely Rad 17 and Rad9-Hus1-Rad1 (9-1-1), induced by ionizing radiation produced		

	DNA damage. These studies utilize both in vitro and in vivo model systems. Sutdies on malignant transformation in vitro utilize HTori-3 cells, which represent a human thyroid epithelial cell model system, and studies on gene expression and function utilize HTori-3 cells and Sprague-Dawley rats as the model systems. Studies with x-rays are also being performed so that Relative Biolgoical Effectiveness (RBE) values for the HZE particles can be determined for radiation induced malignant transformation. By studying the mechanisms by which selenomethionine protects against space radiation induced biological effects, information will be obtained relevant to the mechanisms involved in HZE particle induced radiation carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Knowledge of the mechanisms by which selenomethionine serves as a countermeasure for radiation induced biological effects could help to develop more effective ways to prevent radiation induced biological damage leading to the known biological effects of radiation (e.g., cell and organ toxicity, mutation and malignant transformation). Research on the mechanisms involved in the prevention of radiation induced biological effects will also have earth benefits, as life on Earth involves continual exposure to various types of radiations which are capable of producing adverse biological effects.
Task Progress:	Progress in all of the different types of studies described in the grant proposal has been made in the past year. Studies specifically related to the gene expression area of research are described briefly below. During space travel, astronauts are exposed to higher doses of ionizing radiation than those encountered on earth, which is expected to increase their cancer risk. Of particular concern is the radiation from particles of high atomic number and high energy, known as HZE particles. Prolonged exposure to radiation from HZE particles, as would occur during a mission to Mars, imposes a need for radioprotective agents, which would prevent or lower the risk for the development of cancer. One such agent with promising radioprotective and cancer chemopreventive qualities is L-Selenomethionine (SeM). Our research program is focused on the mechanism(s) by which L-Selenomethionine (SeM) reduces the levels of biological effects associated with the exposure of cells to high-energy, highly charged particles known as HZE particles. The studies explore the mechanisms by which SeM can serve as a countermeasure for radiation effects, such as the induction of cancer. Of particular interest are mechanisms by which SeM may regulate gene expression related to radiation induced carcinogenesis. HTori-3 cells, a human thyroid epithelial cell line, were maintained in the absence or presence of SeM, and subsequently exposed to 0, 10 or 20 cGy of 1.0 GeV/amu iron ions. Gene expression in these cells was analyzed using microarray technology. The signals acquired by scanning the hybridized Gene Chips (Affymetrix) were analyzed using the VMAxS Platform (ViaLogy). Gene expression in irradiated cells, with or without SeM, was compared to non-irradiated samples. Significantly regulated genes were assigned to appropriate pathways using Gene Ontology and KEGG (Kyoto Encyclopedia of Genes and Genomes) Protein Networks. Further, we used EASE over-representation analysis (Douglas A. Hosack, Glynn Dennis Jr., Brad T. Sherman, H. Clifford Lane and Richard A. Lempicki. Genome Biology 2003 4(6):P4) to test for Gene Ontology classes that occurred more frequently in a list of genes of interest (significantly regulated genes) than would be expected by chance given the prevalence of a class among all genes assayed. Our hypothesis is that SeM prevents or suppresses biological effects in cells exposed to HZE particle radiation by altering gene expression patterns in irradiated cells. Our data clearly show the effectiveness of SeM as a countermeasure for HZE particle radiation-induced differential gene expression. For both radiation doses used, a non-toxic 10 cGy and a slightly toxic 20 cGy dose of radiation from iron ions, supplementation with SeM leads to a clear change in the pattern of HZE particle radiation-induced differential gene expression. In HTori-3 cells exposed to a 10 cGy radiation dose from iron ions, 218 genes are up-regulated and 82 genes are down-regulated in the absence of SeM; in cells irradiated with the same dose of radiation in the presence of SeM, only 75 out of 218 up-regulated and 15 out of 82 down-regulated genes are still differentially expressed. In HTori-3 cells exposed to a 20 cGy dose of radiation from iron ions, 176 genes are up-regulated and 124 genes are down-regulated in the absence of SeM; in cells exposed to the same dose of radiation in the presence of SeM, only 53 out of 176 up-regulated and 9 out of 124 down-regulated genes are still differentially expressed. Over-representation analysis of significantly regulated genes indicates that radiation exposure to 10 or 20 cGy radiation doses from iron ions induces up-regulation of genes with activities related to transcription and regulation of transcription, and down-regulation of genes involved in cell growth, mitosis and the cell cycle. In addition, a 10 cGy radiation dose from iron ions up-regulates structural molecules and components of extracellular matrix and cell junctions, while it causes down-regulation of genes involved in synthesis and degradation of proteins. A 20 cGy dose of radiation from iron ions appears to cause up-regulation of genes with roles in cell death and apoptosis, and down-regulation of genes involved in transcription and cell signaling cascades. The results of our analysis of the microarray data appear very promising. The fact that we have detected such a large effect of SeM on HZE radiation-induced differential gene expression is exciting. It will lead us to the next phase of our studies in which we will evaluate changes in the activity of the candidate genes identified through microarray analysis, and attempt to determine the molecular pathways involved in HZE particle radiation and the SeM modification of those effects. It is hoped that the mechanisms by which L-Selenomethionine protects cells from HZE radiation-induced biological effects can be elucidated from these types of studies.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Articles in Peer-reviewed Journals	Kennedy, A.R.; Ware, J.H.; Guan, J.; Donahue, J.J.; Biaglow, J.E.; Zhaozong, Z.; Stewart, J.; Vazquez, M.; Wan, X.S. "Selenomethionine protects against adverse biological effects induced by space radiation." Free Radical Biology & Medicine, Vol. 36, no. 2, 259-266 , Jan-2004